

Lab 03

Name Kaniz Fatema

ID: 20245103154

The Tables Used Here Are from **Bank Database**

1. Find the number of customers from all cities in the **customer** relation

```
SELECT customer_city, count(customer_name) "Total no. of customers" FROM customer  
GROUP BY customer_city;
```

customer_city	Total no. of customers
Brooklyn	1
Harrison	2
Palo Alto	1
Pittsfield	2
Princeton	1
Rye	2
Stamford	2
Woodside	1

2. Find the total number of loans from **Loan** relation from each branch

```
SELECT branch_name, count(loan_number) "No. of loans" FROM loan GROUP BY  
branch_name;
```

branch_name	No. of loans
Downtown	2
Mianus	1
Perryridge	2
Redwood	1
Round Hill	1

3. Find the total amount of loan from **Loan** relation of each branch which amount is greater than 1200

```
SELECT branch_name, SUM(amount) FROM loan GROUP BY branch_name HAVING SUM(amount) > 1200;
```

branch_name	SUM(amount)
Downtown	2500
Perryridge	2800
Redwood	2000

4. Find the average amount from each branch of **loan** relation

```
SELECT branch_name, AVG(amount) "Average amount" FROM loan GROUP BY branch_name;
```

branch_name	Average amount
Downtown	1250.0000
Mianus	500.0000
Perryridge	1400.0000
Redwood	2000.0000
Round Hill	900.0000

5. Find the total amount of each branch from **loan** relation

```
SELECT branch_name, SUM(amount) "Total amount" FROM loan GROUP BY branch_name;
```

Showing rows 0 - 4 (5 total, Query took 0.0002 seconds.)

```
SELECT branch_name, SUM(amount) "Total amount" FROM loan GROUP BY branch_name;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25

Extra options

branch_name	Total amount
Downtown	2500
Mianus	500
Perryridge	2800
Redwood	2000
Round Hill	900

6. Find the total number of tuples for loan and account relation

```
SELECT count(*) FROM loan;
```

Showing rows 0 - 0 (1 total, Query took 0.0001 seconds.)

```
SELECT count(*) FROM loan;
```

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☐ Show all | Number of rows: 25

Extra options

count(*)

7

☐ Show all | Number of rows: 25

7. Find the average account balance of each branch whose average account balance is greater than 500

```
SELECT branch_name, AVG(balance) "Total balance" FROM account GROUP BY branch_name
HAVING AVG(balance) > 500;
```

✓ Showing rows 0 - 2 (3 total, Query took 0.0002 seconds.)

```
SELECT branch_name, AVG(balance) "Total balance" FROM account GROUP BY
branch_name HAVING AVG(balance) > 500;
```

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Extra options

branch_name	Total balance
Brighton	825.0000
Mianus	700.0000
Redwood	700.0000

8. Find the name of all those customers who has either a loan or an account or both

```
(SELECT customer_name FROM borrower) UNION (SELECT customer_name FROM depositor);
```

```
(SELECT customer_name FROM borrower) UNION (SELECT customer_name FROM depositor);
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)☐ Show all | Number of rows: 25 ▼[Extra options](#)**customer_name**

Adams

Curry

Hayes

Johnson

Jones

Smith

Williams

Lindsay

Turner

9. Find the name of all those customers who has both a loan and an account

```
(SELECT customer_name FROM borrower) INTERSECT (SELECT customer_name FROM depositor);
```

```
(SELECT customer_name FROM borrower) INTERSECT (SELECT customer_name FROM depositor);
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)☐ Show all | Number of rows: 25 ▼[Extra options](#)**customer_name**

Hayes

Johnson

Jones

Smith

10. Find the name of all those customers who has only an account but not any loan

```
SELECT customer_name FROM depositor WHERE customer_name NOT IN (SELECT customer_name FROM borrower);
```

```
SELECT customer_name FROM depositor WHERE customer_name NOT IN (SELECT customer_name FROM borrower);
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)☐ Show all | Number of rows: 25 ▼[Extra options](#)**customer_name**

Lindsay

Turner

11. Change the column name from **branch_city** of branch relation to **city**

```
ALTER TABLE branch CHANGE COLUMN branch_city city varchar (32);
```

branch_name	city	assets
Brighton	Brooklyn	7100000
Downtown	Brooklyn	9000000
Mianus	Horseneck	400000
North Town	Rye	3700000
Perryridge	Horseneck	1700000
Pownal	Bennington	300000
Redwood	Palo Alto	2100000
Round Hill	Horseneck	8000000